MATHEMATICS

Applications and interpretation

Higher Level | Set B | Paper 1

Challenge mock examination

Time allowed: 2 hours Maximum mark: 110

Candidate name: _____

Candidate number: _____

Date: _____

Instructions

• A graphic display calculator and a clean Mathematics: applications and interpretation formula booklet are required.

• Answer all questions. Show working and explain mathematical reasoning in context.

• All questions are compulsory. There is no optional section.

• Give exact answers where appropriate. Otherwise give numerical answers to three significant figures, unless a question specifies a different accuracy.

• Keep unrounded values in subsequent calculations. State units where relevant.

• Diagrams are not necessarily drawn to scale. Use the stated data and conditions.

• The number in square brackets gives the maximum mark for a part.

Thriving Scholars original practice | 2026 challenge series

Question 1 | Question paper

Question 1 [Maximum mark: 6]

An account pays a nominal annual interest rate of 4.8%, compounded monthly. Deposits of 800 dollars are made at the end of every month for 10 years. The initial balance is zero. Annual inflation is assumed to remain at 2.5%. Ignore tax and fees.

(a) Write down the monthly interest rate as a decimal. [1]

(b) Find the balance immediately after the 120th deposit. [2]

(c) Express that balance in today's purchasing power. [1]

(d) Find the monthly deposit required to achieve purchasing power of 100 000 dollars after 10 years, under the same assumptions. [2]

Question 2 | Question paper

Question 2 [Maximum mark: 6]

A chemical sensor has response $S(c) = (18c + 12)/(c + 4)$ mV at concentration $c \geq 0$ mg L^{-1}. The response is increasing on this physical domain.

(a) Find c as a function of S, and state the physical domain of this inverse. [3]

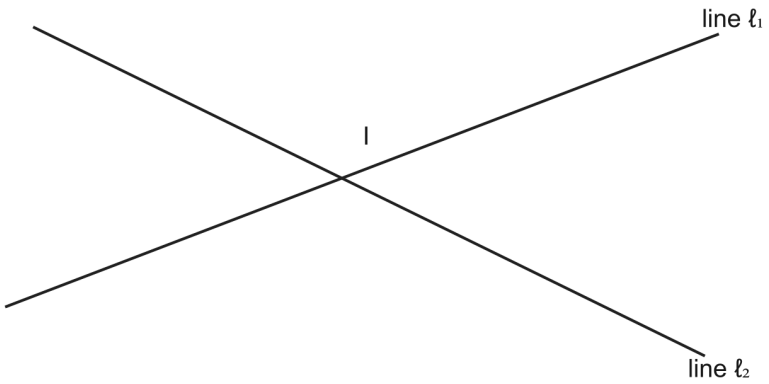
(b) Find the concentration corresponding to 14 mV. [1]

(c) Compare the concentration changes corresponding to signal increases from 13 to 14 mV and from 15 to 16 mV. Explain what this says about sensitivity at high concentration. [2]

Question 3 | Question paper

Question 3 [Maximum mark: 6]

Two maintenance rails are modelled by $\ell_1 : \mathbf{r} = (1, 2, 3) + s(2, -1, 2)$ and $\ell_2 : \mathbf{r} = (5, 0, 7) + t(1, 2, -2)$, with real s, t. Coordinates are in metres. The schematic shows only that the rails intersect.



Schematic intersection; not a coordinate projection

(a) Find their intersection point. [2]

(b) Find the acute angle between the rails. [2]

(c) Find the perpendicular distance from $(1, 2, 3)$ to ℓ_2. [2]

Question 4 | Question paper

Question 4 [Maximum mark: 6]

Seven hikers report their weekly training distance x (km) and the time y (minutes) to complete a steep climb.

Hiker	A	B	C	D	E	F	G
x	12	14	14	18	20	23	25
y	44	40	42	35	37	31	28

Rank both variables in ascending order, using average ranks for ties.

(a) Write down the rank lists for x and y. [2]

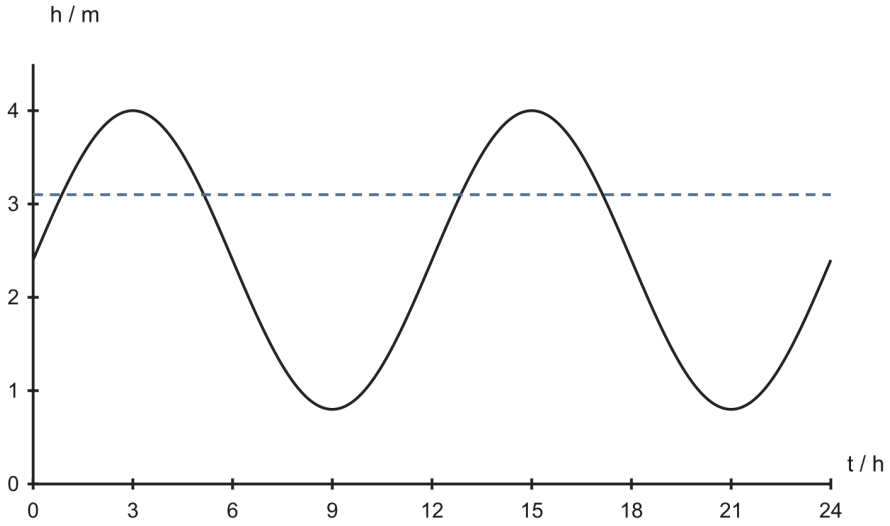
(b) Calculate Spearman's rank correlation coefficient. [2]

(c) Interpret the sign and magnitude of this coefficient. State why it does not by itself prove that training causes a faster climb. [2]

Question 5 | Question paper

Question 5 [Maximum mark: 6]

Tidal depth at a harbour is modelled by $h(t) = 2.4 + 1.6\cos(\pi(t-3)/6)$ metres, for $0 \le t \le 24$, where t is hours after midnight. A survey vessel requires a depth of at least 3.1 m throughout an uninterrupted 90-minute operation. Angles in the formula are in radians.



(a) State the amplitude and period of the tidal model. [2]

(b) Find the first interval after midnight during which the depth is at least 3.1 m. [2]

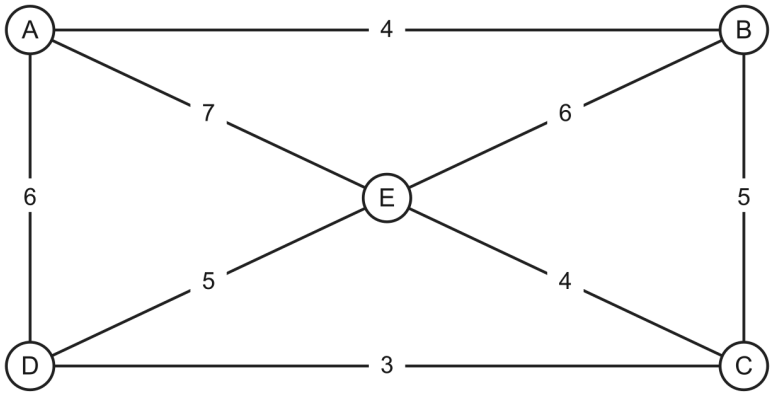
(c) Find the latest start time in this first interval for the operation. Give hours after midnight, without converting to clock time. [2]

Question 6 | Question paper

Question 6 [Maximum mark: 7]

A robot must inspect every corridor in the undirected network at least once and return to its starting station A. Edge labels are corridor lengths in metres. It may repeat corridors; crossings away from labelled stations are not junctions.

Corridor lengths / m



(a) Explain why traversing every corridor exactly once in a closed route is impossible. [1]

(b) Find the minimum total length of corridors that must be repeated. [3]

(c) Find the minimum length of a complete inspection route and explain why your construction permits a closed route. [3]

Question 7 [Maximum mark: 7]

A factory claims that a new process reduces the defect probability from 0.20. Thirty independent items are tested and two are defective. Use a one-tailed exact binomial test at the 5% level. For a lower-tail critical region, choose the largest rejection probability strictly below 0.05.

(a) State the hypotheses for the new defect probability p. [1]

(b) Find the p-value and state the test decision. [2]

(c) Find the critical region, showing that the next larger lower-tail region is not permitted. [2]

(d) Find the probability of a Type II error when the true defect probability is 0.10. [2]

Question 8 [Maximum mark: 7]

Nine independent observations of the lifetime of a component, in hours, are

$42.1, 44.8, 41.7, 46.3, 43.5, 45.2, 42.9, 44.1, 43.7$.

Assume a random sample from a normal population whose variance is unknown.

(a) Find the sample mean and unbiased sample standard deviation. [2]

(b) Find a 95% confidence interval for the population mean lifetime. [3]

(c) Comment on a claimed mean lifetime of 44.5 hours, and explain why the interval is not a range containing 95% of individual lifetimes. [2]

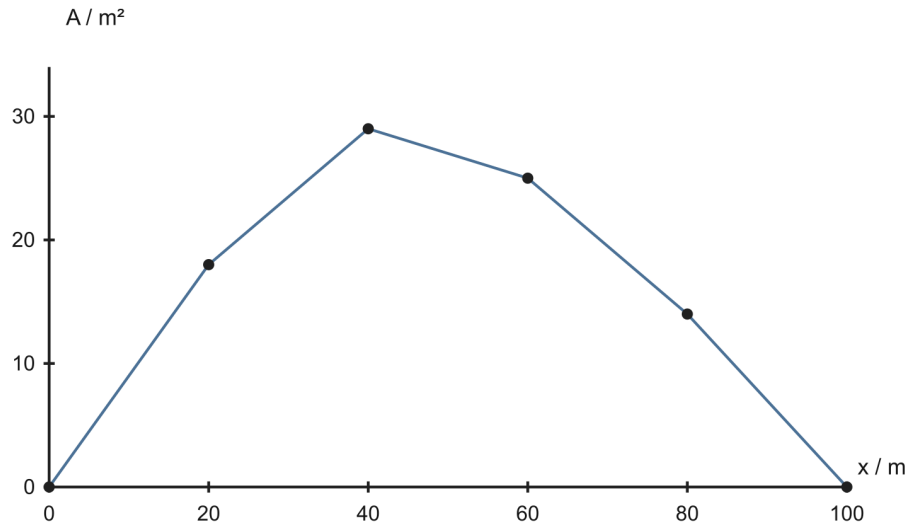
Question 9 | Question paper

Question 9 [Maximum mark: 7]

The cross-sectional area $A(x)$ of a flooded tunnel is measured perpendicular to its axis at equally spaced positions.

x / m	0	20	40	60	80	100
$A(x)$ / m^2	0	18	29	25	14	0

Assume $A(x)$ is continuous and strictly concave down between the ends.



(a) Use the trapezoidal rule with intervals of width 20 m to estimate the water volume. [3]

(b) Additional areas at $x = 10, 30, 50, 70, 90$ are respectively $9.5, 24, 28.5, 20, 7.5$ m^2. Find the trapezoidal estimate using width 10 m. [2]

(c) Using the concavity assumption, state whether these trapezoidal estimates are underestimates or overestimates, and justify your answer. [2]

Question 10 | Question paper

Question 10 [Maximum mark: 7]

A two-stage population has juvenile and adult state vector $\mathbf{s}_n$. It follows $\mathbf{s}_{n+1} = L\mathbf{s}_n$, where

$L = \begin{pmatrix} 0 & 1.8 \\ 0.5 & 0.3 \end{pmatrix}$

. Initially there are 200 juveniles and 100 adults. Fractional population predictions are permitted within the model.

(a) Find the predicted juvenile and adult populations one year later. [2]

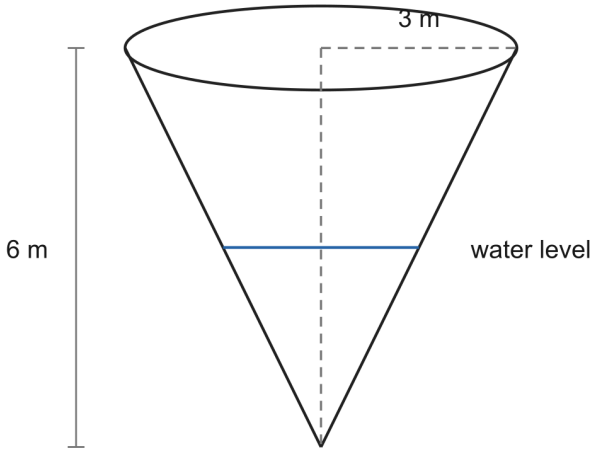
(b) Find the dominant eigenvalue and interpret its long-run growth implication. [2]

(c) Find the long-run juvenile-to-adult ratio and the long-run adult proportion. [3]

Question 11 | Question paper

Question 11 [Maximum mark: 7]

A conical tank has height 6 m and top radius 3 m, with its vertex at the bottom. Water depth is h metres. Water enters at 0.4 $\text{m}^3\ \text{min}^{-1}$ and leaks at $0.02h\ \text{m}^3\ \text{min}^{-1}$.



Conical tank; not to scale

(a) Express the water volume V in terms of h. [2]

(b) Find the rate of increase of the water depth when $h = 2$. [3]

(c) Find the time taken for the depth to rise from 1 m to 2 m, using numerical integration. [2]

Question 12 | Question paper

Question 12 [Maximum mark: 7]

An engineering model relates energy use E (kJ) to load m (kg) by $E = am^b$, with $a > 0$.

m	2	4	6	9	12
E	11.8	25.7	41.4	68.6	95.5

Fit the power model by least-squares linear regression of $\ln E$ on $\ln m$.

(a) Find a and b. [3]

(b) Estimate the energy use at a load of 8 kg. [2]

(c) Show that $(m/E)(dE/dm) = b$. Hence estimate the percentage energy increase caused by a small 5% load increase. [2]

Question 13 [Maximum mark: 7]

During a 15-minute interval, independent arrival counts A and B from two entrances follow Poisson distributions with means 2 and 3 respectively. Let $T = A + B$.

(a) State the distribution of T and find $P(T = 5)$. [2]

(b) Given that five people arrived in total, find the probability that exactly two used entrance A. [3]

(c) Given that nobody used entrance B, find the probability that at least one person used entrance A. Explain the role of independence. [2]

Question 14 [Maximum mark: 8]

A courier must start at A, visit each of B,C,D,E once, and return to A. The complete undirected distance table, in kilometres, is shown.

	A	B	C	D	E
A	—	6	9	10	7
B	6	—	5	8	9
C	9	5	—	6	8
D	10	8	6	—	4
E	7	9	8	4	—

(a) Apply the nearest-neighbour algorithm starting at A. State the resulting tour and an upper bound for the optimum. [3]

(b) Use the deleted-vertex method, deleting A, to obtain a lower bound. [3]

(c) Determine the optimal tour length and justify your conclusion. [2]

Question 15 | Question paper

Question 15 [Maximum mark: 8]

An object initially at $90°\text{C}$ cools according to $dT/dt = -0.15(T - A)$, where t is minutes and A is ambient temperature. For $0 \leq t < 4$, $A = 20$. At $t = 4$ the object is moved instantly to a room with $A = 5$; its own temperature is continuous and the same differential model then applies.

(a) Find the object's temperature at $t = 4$. [2]

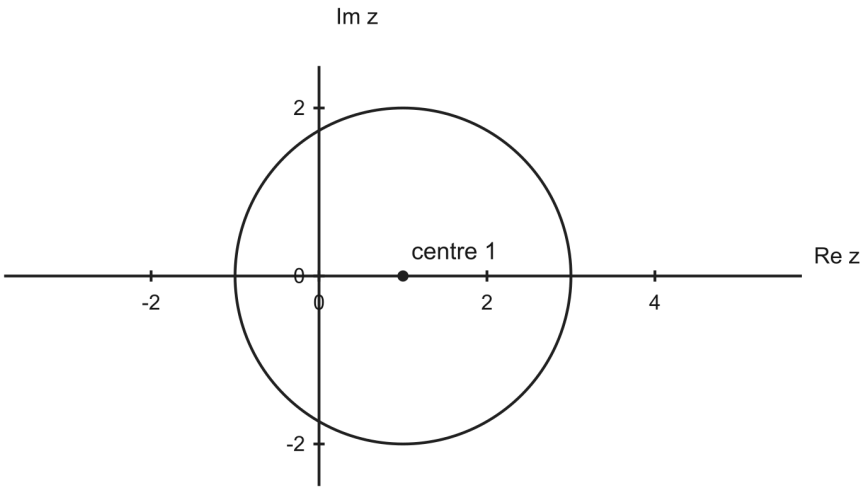
(b) Find the first time after $t = 0$ at which the temperature reaches $15°\text{C}$. [3]

(c) Find the cooling rates immediately before and after the move, and explain why the derivative changes even though temperature is continuous. [3]

Question 16 | Question paper

Question 16 [Maximum mark: 8]

Points on a circle are described by $z = 1 + 2e^{it}$, $0 \leq t < 2\pi$. A transformation is $w = (0.6 + 0.8i)z + 2 - i$. The diagram shows the original circle in the z-plane. Angles are in radians.



(a) Describe the rotation and scale factor associated with multiplication by $0.6 + 0.8i$. [2]

(b) Find the centre and radius of the image circle in the w-plane. [3]

(c) Find the value of t for which $\text{Im}\, w$ is greatest, and find that maximum imaginary part. [3]

End of question paper

MATHEMATICS

Applications and interpretation

Higher Level | Set B | Paper 1

Worked solution book

Complete this paper before consulting the worked solutions. Each part has method (M), accuracy (A) and reasoning (R) points. The number following a code is its mark value. Award only the points supported by the candidate's retained working.

Accept mathematically equivalent valid methods. Numerical answers may use technology, but should be supported by equations, test definitions, bounds, labelled graphs or other appropriate mathematical working. Retain unrounded intermediate values.

Use the portal to enter a whole-number self-assessed mark for each part. These are independent practice papers; the self-assessed report does not calculate an official IB grade.

Question 1 [Maximum mark: 6]

(a) [1 marks]

A1 $i = 0.048/12 = 0.004$.

(b) [2 marks]

M1 Use the ordinary-annuity future value $800[(1.004)^{120} - 1]/0.004$.

A1 The balance is 122905.57 dollars.

(c) [1 marks]

A1 Divide by 1.025^{10}, giving 96013.633 dollars in today's money.

(d) [2 marks]

M1 Set the target future balance to $100000(1.025)^{10}$ and solve for the deposit.

A1 The required deposit is 833.21501 dollars per month (round up to the next cent if an attainable minimum payment is required).

Question 2 | Worked solutions

Question 2 [Maximum mark: 6]

(a) [3 marks]

M1 Rearrange $Sc + 4S = 18c + 12$.

A1 $c = (4S - 12)/(18 - S)$.

A1 The inverse domain is $3 \le S < 18$, since $S(0) = 3$ and the response approaches 18 from below.

(b) [1 marks]

A1 $c = (56 - 12)/4 = 11$ mg L^{-1}.

(c) [2 marks]

A1 The concentration changes are $11 - 8 = 3$ and $26 - 16 = 10$ mg L^{-1} respectively.

R1 A larger concentration change gives the same 1 mV response change at high concentration; the sensor is less sensitive there.

Question 3 | Worked solutions

Question 3 [Maximum mark: 6]

(a) [2 marks]

M1 Equate the three components; $s = 2, t = 0$ satisfies all of them.

A1 The intersection is $(5, 0, 7)$.

(b) [2 marks]

M1 $\cos\theta = |\mathbf{u} \cdot \mathbf{v}|/(|\mathbf{u}||\mathbf{v}|) = 4/9$.

A1 $\theta = 63.6122°$.

(c) [2 marks]

M1 Use $|\overrightarrow{IP} \times \mathbf{v}|/|\mathbf{v}|$, or minimize the squared point-to-line distance.

A1 The distance is 5.3748385 m.

Question 4 | Worked solutions

Question 4 [Maximum mark: 6]

(a) [2 marks]

A1 The x ranks are $[1, 2.5, 2.5, 4, 5, 6, 7]$.

A1 The y ranks are $[7, 5, 6, 3, 4, 2, 1]$.

(b) [2 marks]

M1 Calculate the Pearson correlation of the rank lists, which handles the tied ranks.

A1 $r_s = -0.95499371$.

(c) [2 marks]

R1 There is a strong negative monotonic association: greater training distance tends to be associated with shorter climb times.

R1 The data are observational and may be confounded, for example by baseline fitness or age; correlation is not proof of causation.

Question 5 [Maximum mark: 6]

(a) [2 marks]

A1 Amplitude 1.6 m.

A1 Period 12 hours.

(b) [2 marks]

M1 Solve $\cos(\pi(t-3)/6) \geq 0.4375$ around the first maximum.

A1 The interval is $0.86481599 \leq t \leq 5.135184$ hours.

(c) [2 marks]

M1 Subtract 1.5 hours from the end of the interval.

A1 The latest start is $t = 3.635184$ hours.

Question 6 [Maximum mark: 7]

(a) [1 marks]

R1 Vertices A,B,C,D have odd degree 3. A connected graph admits an Euler circuit only when all degrees are even.

(b) [3 marks]

M1 Pair the four odd vertices using shortest path distances.

A1 The three pairing totals are $7, 17, 11$ m.

A1 Repeat AB and CD, for 7 m in total.

(c) [3 marks]

M1 The original edge total is 40 m.

A1 The minimum route length is $40 + 7 = 47$ m.

R1 Duplicating the selected paths makes every vertex even while preserving connectivity, so an Euler circuit of that length exists.

Question 7 | Worked solutions

Question 7 [Maximum mark: 7]

(a) [1 marks]

A1 $H_0 : p = 0.20, H_1 : p < 0.20.$

(b) [2 marks]

M1 $P(X \leq 2)$ for $X \sim B(30, 0.20)$.

A1 $p = 0.044178985 < 0.05$; reject the null hypothesis in favour of a reduction.

(c) [2 marks]

A1 Reject when $X \leq 2$; its probability is 0.044178985.

R1 $P_0(X \leq 3) = 0.12271081 > 0.05$, so the region cannot be enlarged.

(d) [2 marks]

M1 Failing to reject means $X > 2$, now with $X \sim B(30, 0.10)$.

A1 The Type II error probability is 0.58864876.

Question 8 [Maximum mark: 7]

(a) [2 marks]

A1 $\bar{x} = 43.811111$ h.

A1 $s = 1.4793955$ h.

(b) [3 marks]

M1 Use a t interval with $df = 8$, not the population standard deviation.

M1 $\bar{x} \pm t_{0.975,8} s/\sqrt{9}$, where $t_{0.975,8} = 2.3060041$.

A1 The interval is $(42.673947, 44.948275)$ hours.

(c) [2 marks]

R1 44.5 hours lies inside this interval, so this sample is compatible with that mean at this confidence level.

R1 The interval estimates the population mean using the standard error; it is not a prediction interval for individual observations.

Question 9 [Maximum mark: 7]

(a) [3 marks]

M1 Use $V \approx (20/2)[A_0 + 2(A_1 + A_2 + A_3 + A_4) + A_5]$.

M1 Substitute $18 + 29 + 25 + 14 = 86$.

A1 $V \approx 1720.0\ \text{m}^3$.

(b) [2 marks]

M1 Interleave all eleven equally spaced observations; endpoints are zero.

A1 The refined estimate is $1755.0\ \text{m}^3$.

(c) [2 marks]

R1 Each chord lies below a concave-down graph.

R1 The trapezoids therefore underestimate the integral and hence the volume; the larger refined estimate is consistent with this.

Question 10 | Worked solutions

Question 10 [Maximum mark: 7]

(a) [2 marks]

M1 Multiply $L(200, 100)^T$.

A1 The next state is

$$\begin{pmatrix} 180 \\ 130 \end{pmatrix}$$

(b) [2 marks]

A1 The eigenvalues are 1.1104686 and -0.81046864; the dominant magnitude is 1.1104686.

R1 The model eventually grows by a factor 1.1104686 per year, about 11.05% annually.

(c) [3 marks]

M1 For an eigenvector $(J, A)^T$, $1.8A = \lambda J$.

A1 $J : A = 1.6209373 : 1$.

A1 The adult proportion is $1/(1 + 1.6209373) = 0.3815429$.

Question 11 [Maximum mark: 7]

(a) [2 marks]

M1 Similar triangles give $r = h/2$.

A1 $V = \frac{1}{3}\pi(h/2)^2 h = \pi h^3/12$.

(b) [3 marks]

M1 $dV/dh = \pi h^2/4$.

M1 At $h = 2$, $(\pi h^2/4)(dh/dt) = 0.4 - 0.02h = 0.36$.

A1 $dh/dt = 0.11459156$ m min^{-1}.

(c) [2 marks]

M1 $\Delta t = \int_1^2 \frac{\pi h^2}{4(0.4-0.02h)}\, dh$.

A1 The time is 4.9829001 minutes.

Question 12 | Worked solutions

Question 12 [Maximum mark: 7]

(a) [3 marks]

M1 Linearize as $\ln E = \ln a + b \ln m$.

A1 $b = 1.1721178$.

A1 $a = 5.1549397$.

(b) [2 marks]

M1 Use $5.1549397(8)^{1.1721178}$ with unrounded fitted parameters.

A1 $E = 58.98634\,\text{kJ}$.

(c) [2 marks]

M1 $dE/dm = abm^{b-1}$, so $(m/E)(dE/dm) = b$.

A1 The first-order estimated increase is $5b = 5.8605891\%$. This is a local approximation.

Question 13 [Maximum mark: 7]

(a) [2 marks]

A1 $T \sim \text{Po}(5)$.

A1 $P(T = 5) = e^{-5}5^5/5! = 0.17546737$.

(b) [3 marks]

M1 The numerator is $P(A = 2, B = 3) = P(A = 2)P(B = 3)$.

M1 Divide that joint probability by $P(T = 5)$.

A1 The conditional probability is 0.3456.

(c) [2 marks]

R1 Independence means conditioning on $B = 0$ does not change the distribution of A.

A1 $P(A \geq 1 \mid B = 0) = 1 - e^{-2} = 0.86466472$.

Question 14 [Maximum mark: 8]

(a) [3 marks]

M1 Choose the closest unvisited vertex at each step.

A1 The tour is A-B-C-D-E-A.

A1 Its length is $6 + 5 + 6 + 4 + 7 = 28$ km, an upper bound.

(b) [3 marks]

M1 A minimum spanning tree on B,C,D,E uses DE,BC,CD.

A1 That tree has weight 15 km.

A1 Add the two smallest A-edges, 6 and 7, to obtain a lower bound of 28 km.

(c) [2 marks]

R1 The lower and upper bounds are equal.

A1 The optimum is exactly 28 km; the tour in part (a) attains it.

Question 15 [Maximum mark: 8]

(a) [2 marks]

M1 The first-stage solution is $T = 20 + 70e^{-0.15t}$.

A1 $T(4) = 58.416815^{\circ}\text{C}$.

(b) [3 marks]

M1 For $t \geq 4$, $T = 5 + (58.416815 - 5)e^{-0.15(t-4)}$, retaining the unrounded $T(4)$.

M1 Solve $10 = (T(4) - 5)e^{-0.15(t-4)}$.

A1 The first time is $t = 15.17027$ minutes.

(c) [3 marks]

A1 The rates are -5.7625222 and $-8.0125222^{\circ}\text{C min}^{-1}$ respectively.

M1 The ambient temperature changes by 15 degrees, changing the derivative by $-0.15(15) = -2.25$ degrees per minute.

R1 The driving temperature difference changes immediately, while the object's temperature does not jump.

Question 16 | Worked solutions

Question 16 [Maximum mark: 8]

(a) [2 marks]

A1 The scale factor is $\sqrt{0.6^2 + 0.8^2} = 1$.

A1 The rotation is anticlockwise through 0.92729522 rad about the origin.

(b) [3 marks]

M1 Transform the original centre $z = 1$ and multiply the radius by $|0.6 + 0.8i|$.

A1 The centre is $2.6 - 0.2i$.

A1 The radius remains 2.

(c) [3 marks]

M1 $w = 2.6 - 0.2i + 2e^{i(t + \arg(0.6 + 0.8i))}$.

A1 The maximum occurs at $t = 0.64350111$ rad, where the rotating radius points upwards.

A1 The maximum imaginary part is $-0.2 + 2 = 1.8$.

End of worked solution book